

Frame Description

In Auto-Inventory mode, the response frame we will receive is explained below:

- <SOF> "Starting Of Frame" – **1 byte**
- <FRAME> "Covering Command code, Flags & Data"
- <CRC> "Cyclic Redundancy Check" – **2 bytes**.

Now let me split the <FRAME> for more clarity:

- <Len> "Frame Length" – **1 byte**
- <0xE0> "Fixed 1 byte before Flags information" – **1 byte**
- <Flags> "say for Device ID, EPC, TID & User Memory" – **1 byte***
- <Device ID> "say Serial No of the Reader/Device" – **4 bytes in REVERSE**
- <EPC Len> "Length of EPC Value" – **1 byte**
- <EPC> "Value of EPC" – **default 12 bytes, but can be changed**
- <TID Len> "Length of TID" – **1 byte**
- <TID> "This is Unique ID" – **12 bytes**
- <Memory Len> "Length of User Memory Configured/Requested" – **1 byte**
- <Memory> "Value from Memory Blocks" – **variable bytes, based on Length Configured**
- <RSSI> "1st one byte for RSSI-H & next byte for RSSI-L" – **2 bytes**.

* Let's understand the 1 byte of <Flags> based on our initial Reader/Device configuration...

This 1 byte value is all together calculated value based on our configuration we wish to have in our Frame (i.e., Device ID or EPC or TID or User Memory or Combination of some or All).

Device ID	–	0x40
EPC	–	0x01
TID	–	0x02
Memory	–	0x08
RSSI	–	0x04

So, if we select/configured All to have in our Response Frame; then the 1 byte of <Flags> will be **"4F"**.

Now, if we select/configured only Device ID, EPC & TID; then the 1 byte of <Flags> will be **"43"**...